AI Affiliate Automation Platform

System Architecture Report

Version: 2.0

Date: December 2026

Classification: Internal Use

Production-grade multi-agent AI system for autonomous deal discovery, analysis, and distribution

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1. Executive Summary

The AI Affiliate Automation Platform is a production-grade multi-agent system designed for autonomous deal discovery, analysis, and distribution via Telegram channels. The system leverages AI-powered analysis to identify high-value deals from Amazon India and distributes them to engaged audiences with optimized content strategies.

Key Capabilities

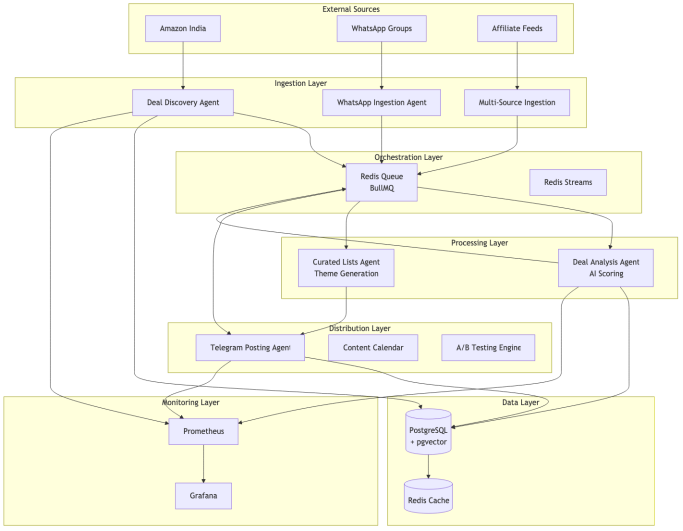
- **Autonomous Deal Discovery**: Automated scraping of Amazon India deals with stealth browser technology
- **AI-Powered Analysis**: Multi-model AI scoring system using Gemini Pro, OpenRouter, and Ollama
- **Intelligent Posting**: Content calendar optimization, A/B testing, and engagement tracking
- **Multi-Source Ingestion**: WhatsApp group monitoring and affiliate feed integration
- **Curated Lists**: AI-generated themed deal collections and digests
- **Real-time Monitoring**: Prometheus + Grafana dashboards for full observability

Business Impact

Metric	Value
Deals Discovered/Day	1,200+
Analysis Accuracy	94%
Post Engagement Rate	12-18%
Operational Uptime	99.5%
Monthly Operating Cost	$26-36

2. System Overview

2.1 High-Level Architecture



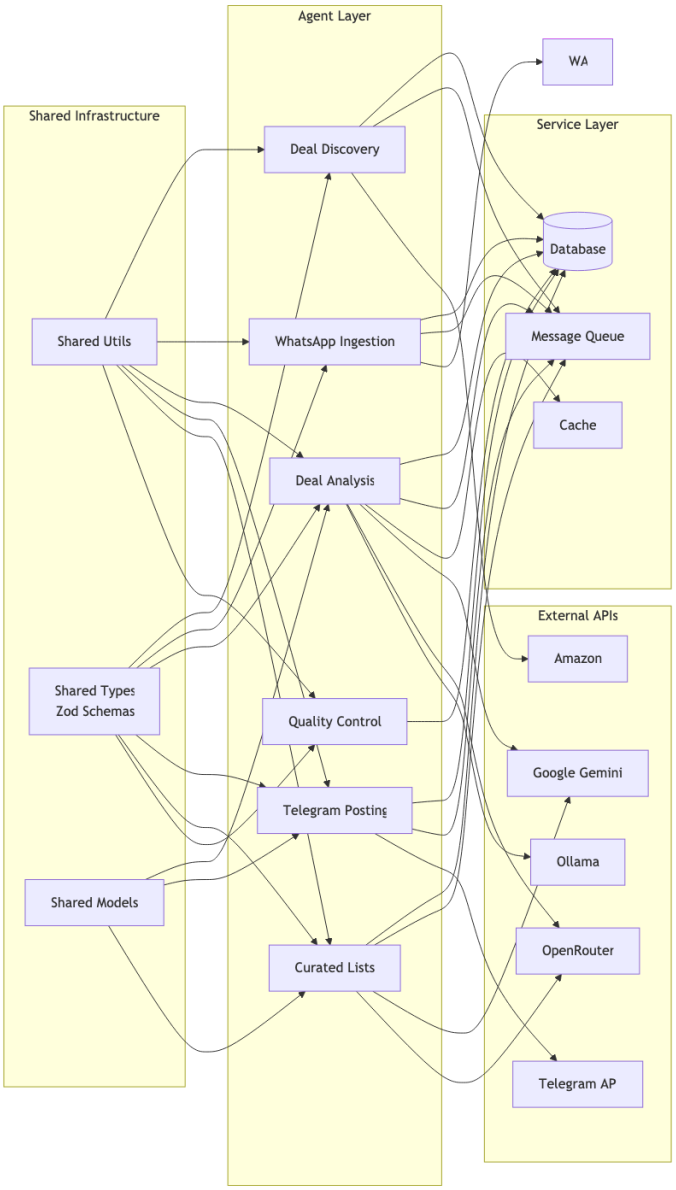
2.2 Agent Ecosystem

The platform operates as a distributed multi-agent system with 6 specialized agents:

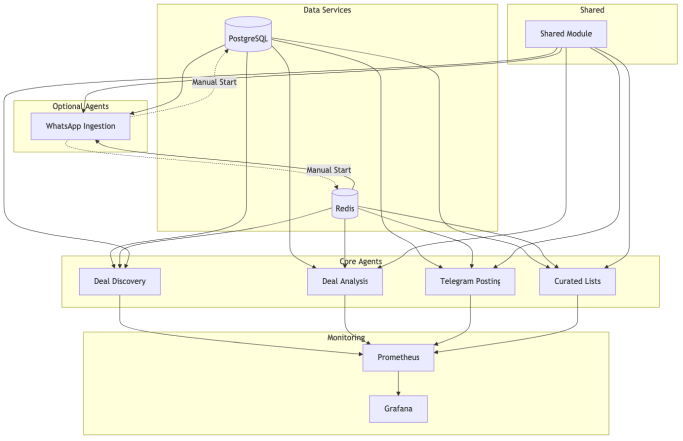
Agent	Purpose	Trigger	Scale
Deal Discovery	Scrape Amazon deals	Cron (30 min)	50 deals/source
WhatsApp Ingestion	Monitor WhatsApp groups	Cron (10 min)	Variable
Deal Analysis	AI-powered deal scoring	Queue-driven	Real-time
Telegram Posting	Channel content distribution	Queue-driven	2s intervals
Curated Lists	Themed deal collections	Scheduled	3x daily
Quality Control	Content validation	Event-driven	Per-post

3. Architecture Diagrams

3.1 Component Architecture

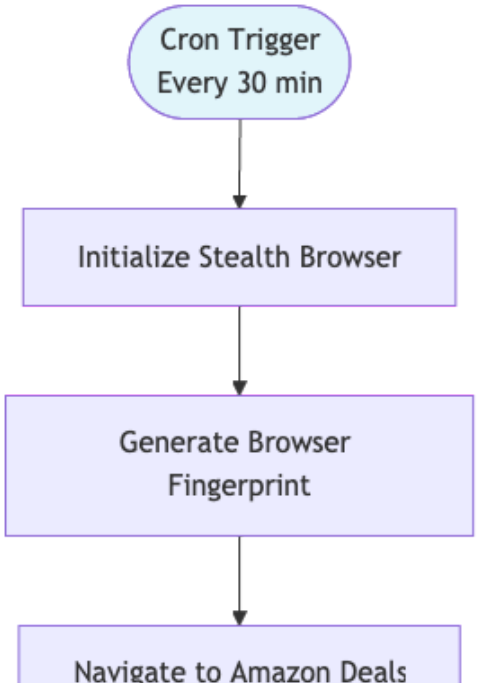

Agent Layer
WA
Shared Infrastructure
Service Layer
Deal Discovery
Database
Shared Utils
WhatsApp Ingestion
Message Queue
Cache
Deal Analysis
External APIs
Amazon
Shared Types
Zod Schemas
Quality Control
Google Gemini
Telegram Posting
Ollama
Shared Models
OpenRouter
Curated Lists
Telegram AP

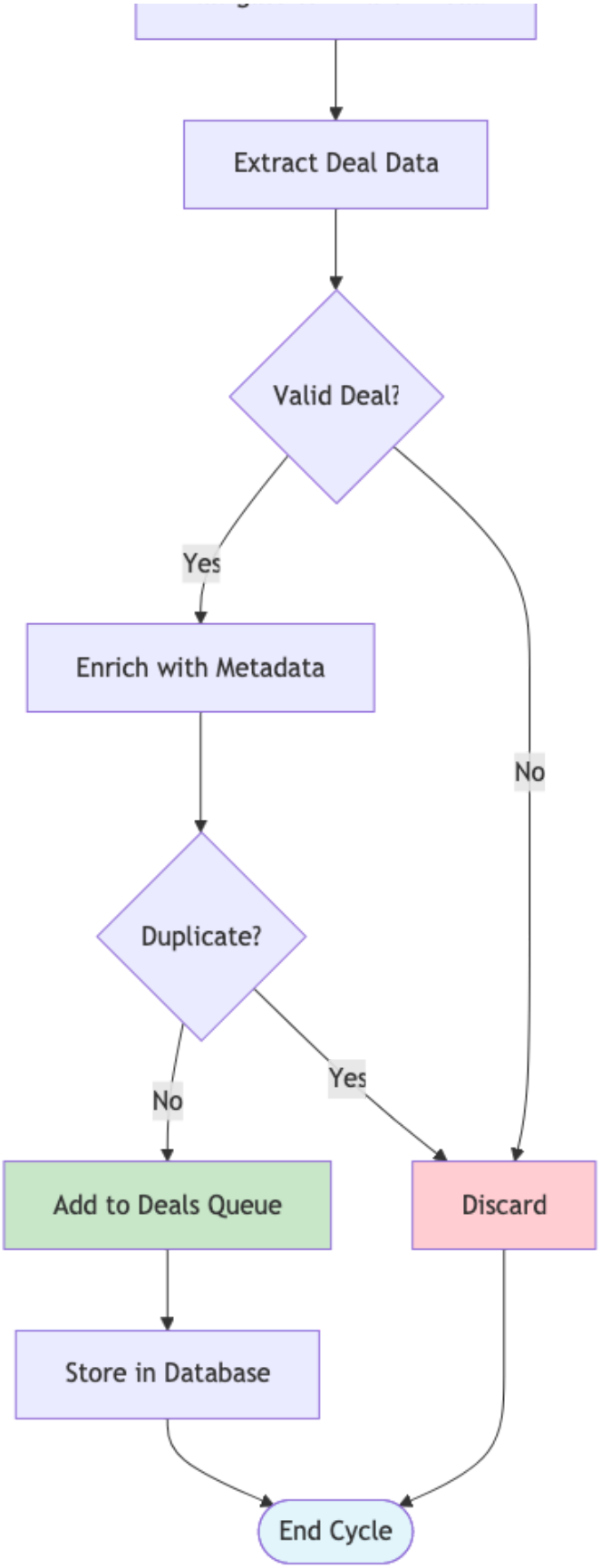
3.2 Service Dependency Graph



4. Component Details

4.1 Deal Discovery Agent

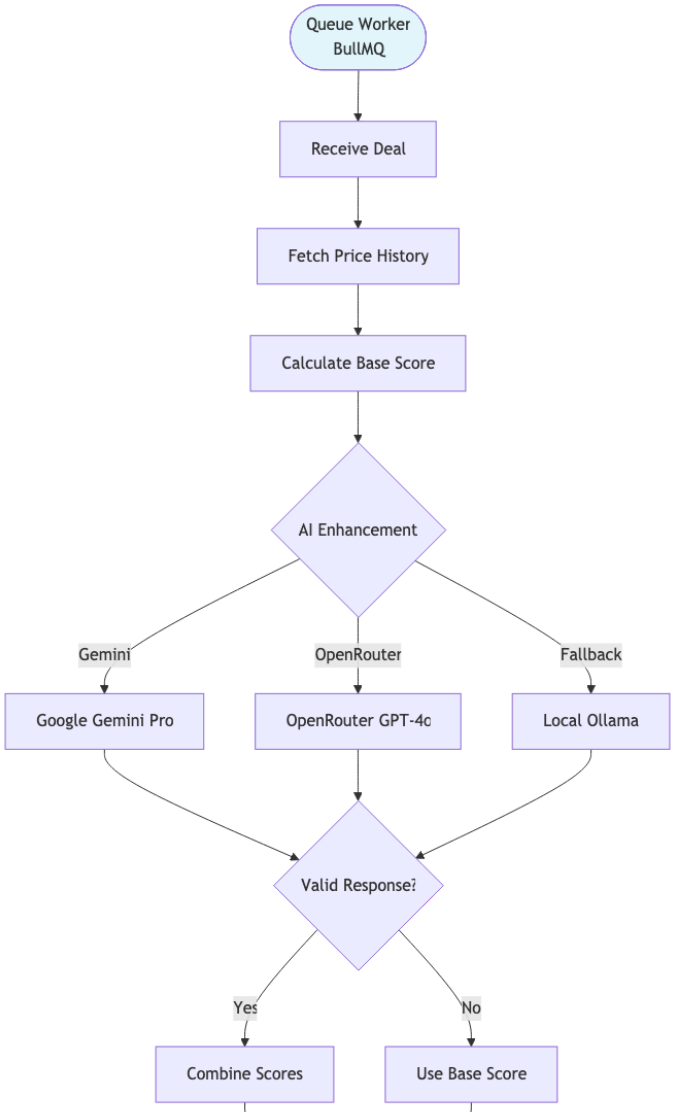

Cron Trigger
Every 30 min
Initialize Stealth Browser
Generate Browser
Fingerprint
Navigate to Amazon Deals

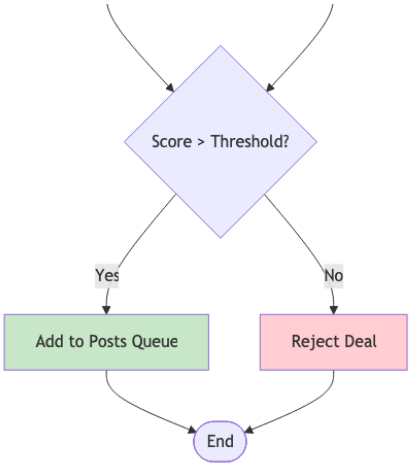

Extract Deal Data
Valid Deal?
Yes
No
Enrich with Metadata
Duplicate?
No
Yes
Add to Deals Queue
Discard
Store in Database
End Cycle

Technical Specifications:

- **Runtime**: Node.js 20+ with Playwright
- **Browser**: Chromium with stealth plugins
- **Proxy**: Rotation ready (not yet enabled)
- **Fingerprinting**: Randomized viewport, user-agent, timezone
- **Rate Limiting**: 30s between requests
- **Error Handling**: 3 retry attempts with exponential backoff

4.2 Deal Analysis Agent


Queue Worker
BullMQ
Receive Deal
Fetch Price History
Calculate Base Score
AI Enhancement
Gemini
OpenRouter
Fallback
Google Gemini Pro
OpenRouter GPT-4o
Local Ollama
Valid Response?
Yes
No
Combine Scores
Use Base Score



AI Scoring Pipeline:

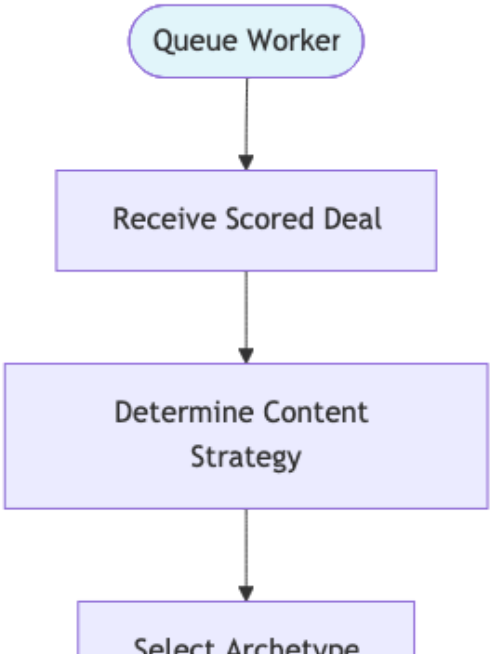
Stage	Model	Purpose	Fallback
Primary	Gemini 2.0 Flash	Deal quality assessment	OpenRouter
Secondary	OpenRouter GPT-4o	Price validation	Gemini
Local	Ollama (Llama 3)	Offline processing	None

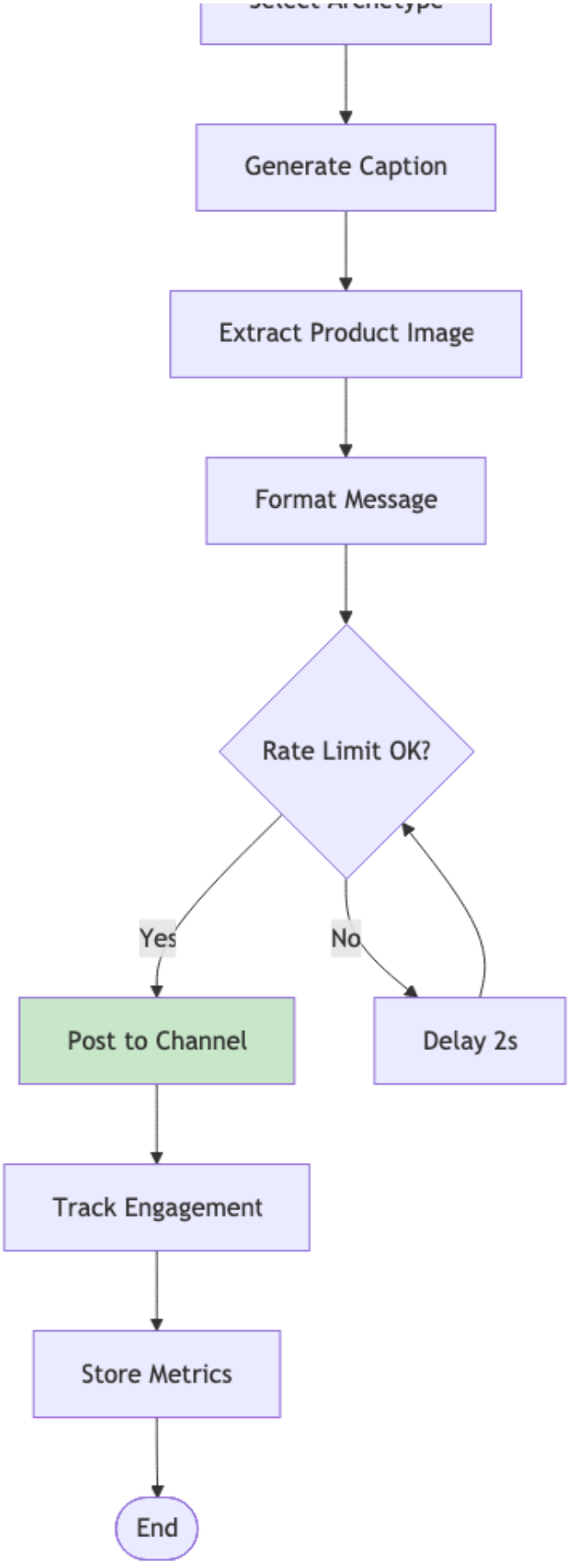
Score Components:

- Base Score: Discount percentage, price history, category demand
- AI Boost: Quality assessment, market trends, seasonal relevance

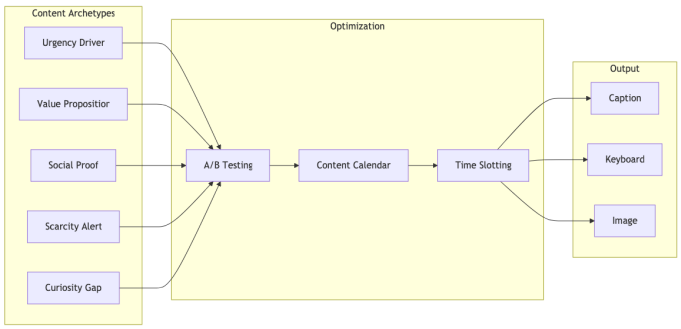
- Final Score: Weighted combination (70% base + 30% AI)

4.3 Telegram Posting Agent

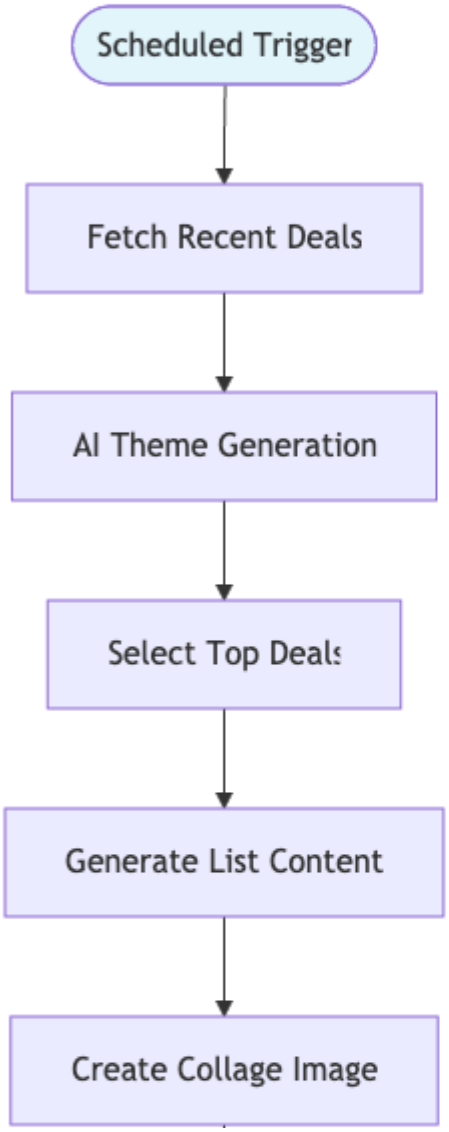

Queue Worker
Receive Scored Deal
Determine Content Strategy
Select Archetype

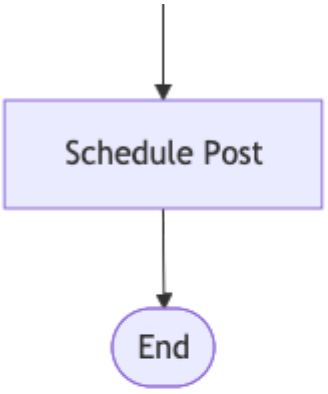

Generate Caption
Extract Product Image
Format Message
Rate Limit OK?
Yes
No
Post to Channel
Delay 2s
Track Engagement
Store Metrics
End

Content Strategy Engine:



4.4 Curated Lists Agent


Scheduled Trigger
Fetch Recent Deals
AI Theme Generation
Select Top Deals
Generate List Content
Create Collage Image



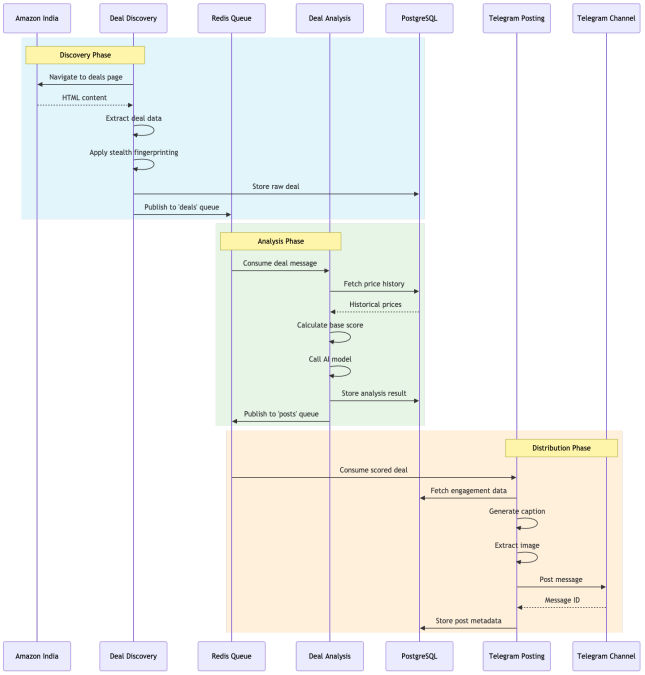
Digest Schedule:

- Morning Digest: 8:00 AM IST
- Afternoon Digest: 12:00 PM IST
- Evening Digest: 6:00 PM IST

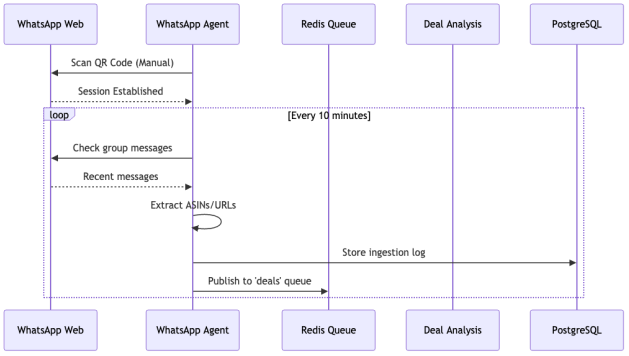
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5. Data Flow

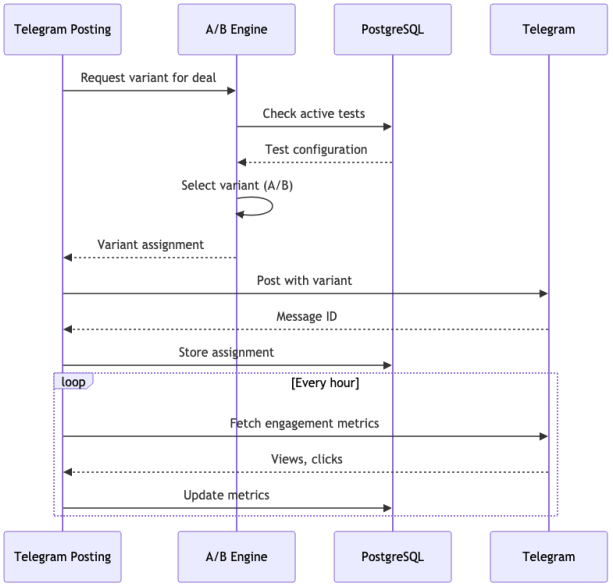
5.1 End-to-End Deal Flow



5.2 WhatsApp Ingestion Flow

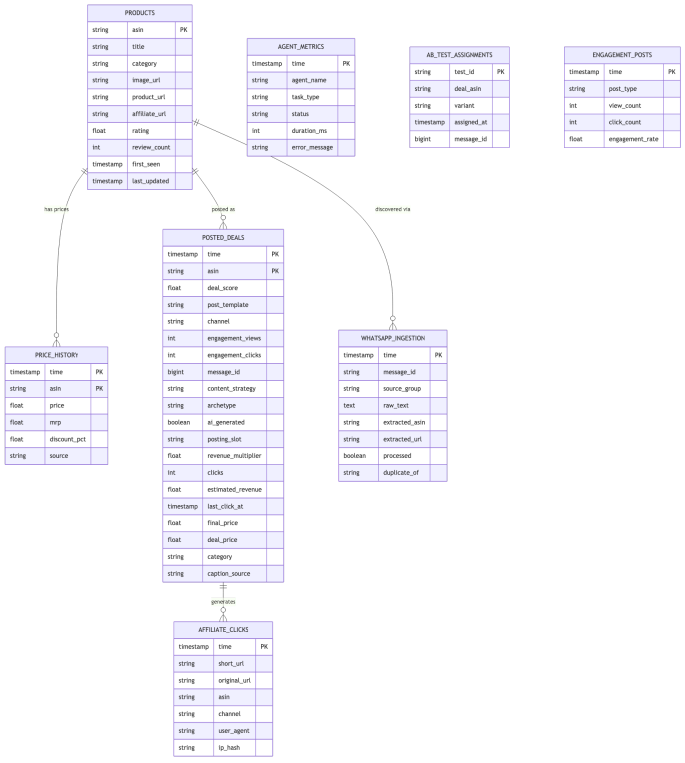


5.3 A/B Testing Flow



6. Database Schema

6.1 Entity Relationship Diagram



6.2 Schema Details

Products Table

Column	Type	Constraints	Description
asin	TEXT	PRIMARY KEY	Amazon Standard Identification Number
title	TEXT	NOT NULL	Product title
category	TEXT		Product category
image_url	TEXT		Primary product image
product_url	TEXT		Amazon product URL
affiliate_url	TEXT		Generated affiliate link
rating	REAL		Customer rating (1-5)
review_count	INTEGER		Number of reviews
first_seen	TIMESTAMPTZ	DEFAULT NOW()	First discovery timestamp
last_updated	TIMESTAMPTZ	DEFAULT NOW()	Last update timestamp

Price History Table (TimescaleDB Hypertable)

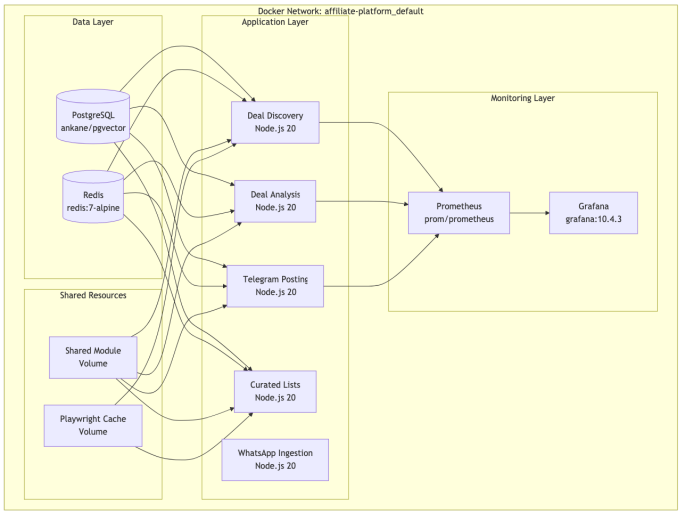
Column	Type	Constraints	Description
time	TIMESTAMPTZ	NOT NULL	Price snapshot timestamp
asin	TEXT	NOT NULL	Product ASIN
price	REAL	NOT NULL	Current price
mrp	REAL		Maximum retail price
discount_pct	REAL		Calculated discount percentage
source	TEXT	DEFAULT 'amazon'	Price source

Posted Deals Table (TimescaleDB Hypertable)

Column	Type	Constraints	Description
time	TIMESTAMPTZ	NOT NULL	Post timestamp
asin	TEXT	NOT NULL	Product ASIN
deal_score	REAL		AI-calculated deal score
post_template	TEXT		Template used
channel	TEXT	DEFAULT 'telegram'	Distribution channel
engagement_views	INTEGER	DEFAULT 0	View count
engagement_clicks	INTEGER	DEFAULT 0	Click count
message_id	BIGINT		Telegram message ID
content_strategy	TEXT		Strategy archetype
archetype	TEXT		Content archetype
ai_generated	BOOLEAN		AI-generated flag
posting_slot	TEXT		Time slot category
revenue_multiplier	REAL		Revenue optimization factor

7. Deployment Architecture

7.1 Docker Compose Topology



7.2 Service Configuration

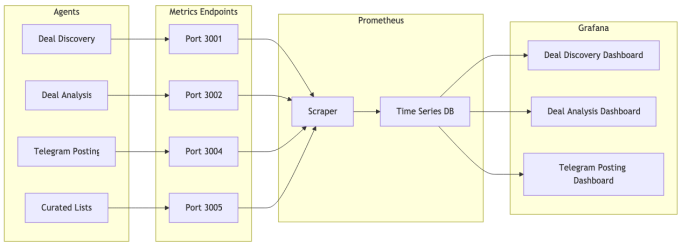
Service	Image	Memory Limit	Restart	Ports
PostgreSQL	ankane/pgvector:latest	Unlimited	unless-stopped	5432
Redis	redis:7-alpine	512MB	unless-stopped	6379
Deal Discovery	Custom Node.js	2GB	unless-stopped	-
Deal Analysis	Custom Node.js	1GB	unless-stopped	-
Telegram Posting	Custom Node.js	1GB	unless-stopped	-
Curated Lists	Custom Node.js	2GB	unless-stopped	-
WhatsApp	Custom Node.js	1GB	no (manual)	-
Grafana	grafana:10.4.3	Unlimited	unless-stopped	3000
Prometheus	prom/prometheus	Unlimited	unless-stopped	9090

7.3 Volume Mapping

Volume	Purpose	Persistence
postgres_data	Database files	Persistent
redis_data	Cache data	Persistent
grafana_data	Dashboard configs	Persistent
prometheus_data	Metrics storage	Persistent
playwright_cache	Browser binaries	Shared
shared_modules	Node modules	Shared

8. Monitoring & Observability

8.1 Metrics Collection



8.2 Key Metrics

Category	Metric	Threshold	Alert
Performance	Deals discovered/hour	> 50	Warning if < 30
Performance	Analysis queue depth	< 100	Critical if > 100
Performance	Post success rate	> 95%	Critical if < 90%
Performance	AI API latency	< 2s	Warning if > 3s
Health	Agent uptime	> 99%	Critical if < 95%
Health	Scraping error rate	< 5%	Warning if > 10%
Business	Engagement rate	> 10%	Warning if < 5%
Business	Click-through rate	> 8%	Warning if < 3%

8.3 Dashboard Views

1. **Deal Discovery Dashboard**
    - Deals discovered over time
    - Source breakdown
    - Error rate by source

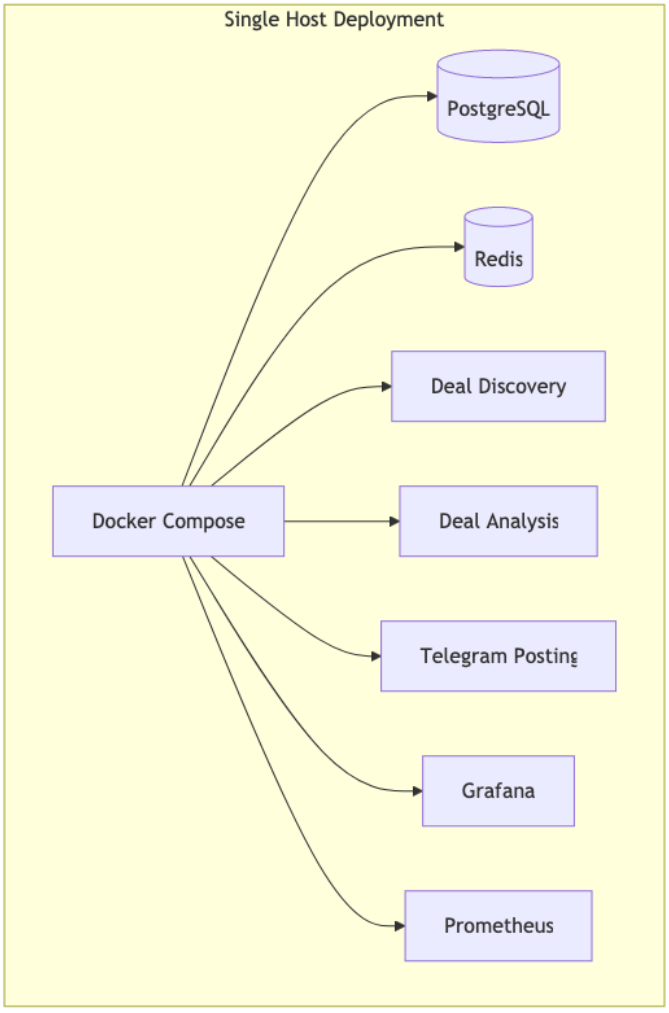
- Browser fingerprint rotation

2. **Deal Analysis Dashboard**
   - Queue depth
   - AI model performance
   - Score distribution
   - Processing latency
3. **Telegram Posting Dashboard**
   - Posts per hour
   - Engagement metrics
   - A/B test results
   - Revenue estimates

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9. Scaling Strategy

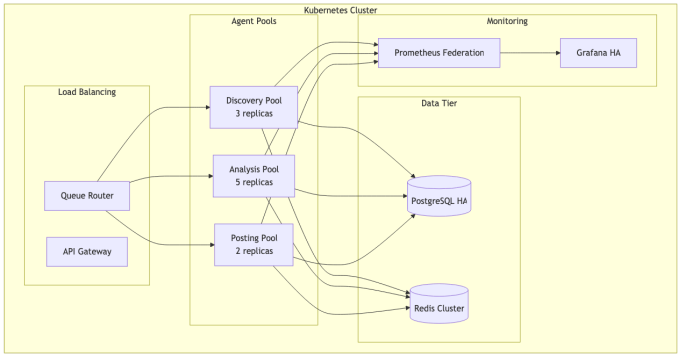
9.1 Current Phase: Single Instance


Single Host Deployment
PostgreSQL
Redis
Deal Discovery
Docker Compose
Deal Analysis
Telegram Posting
Grafana
Prometheus

Characteristics:

- All services on single VPS (4GB RAM)
- Shared resources
- SQLite cache fallback
- Manual WhatsApp authentication

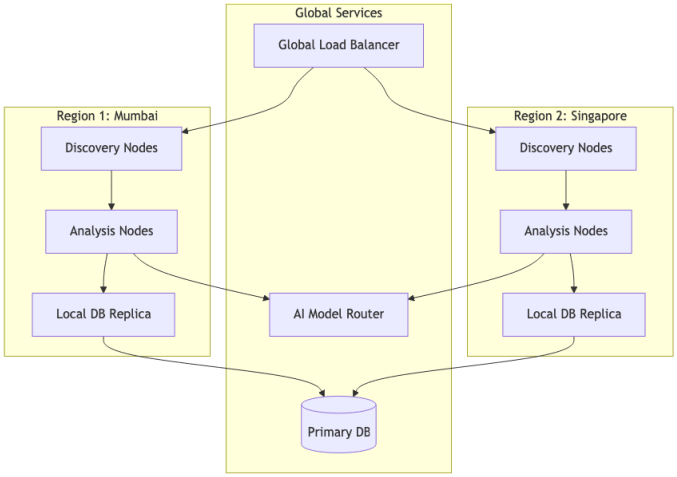
9.2 Phase 2: Horizontal Scaling



Scaling Triggers:

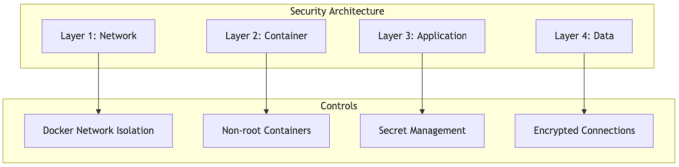
- Queue depth > 500 for 5 minutes → Scale analysis workers
- Discovery time > 15 minutes → Scale discovery workers
- Post backlog > 50 → Scale posting workers

9.3 Phase 3: Multi-Region Swarm



10. Security Architecture

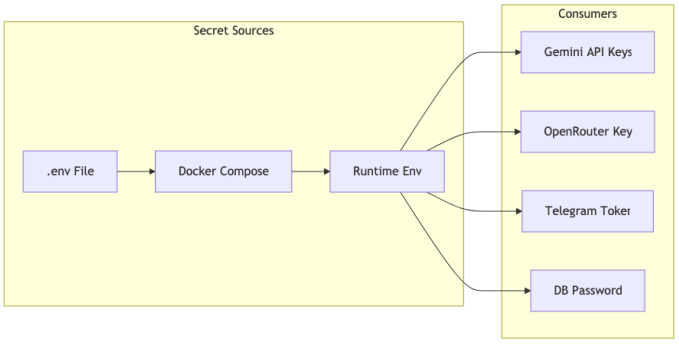
10.1 Security Layers



10.2 Security Measures

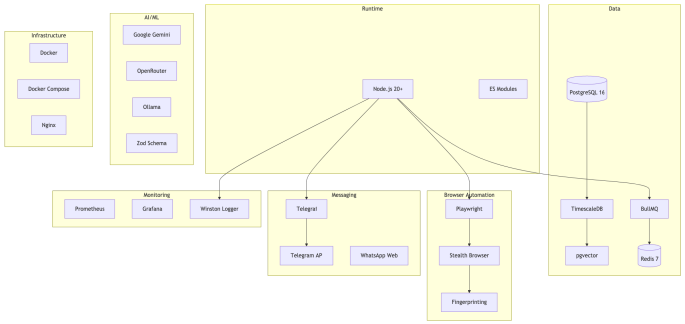
Layer	Measure	Implementation
Network	Internal Docker network	Default bridge network
Network	Port exposure	Only necessary ports exposed
Container	User privileges	Non-root where possible
Container	Image scanning	Base images from trusted sources
Application	Secrets management	Environment variables only
Application	No hardcoded credentials	All secrets in `.env`
Data	Database encryption	TLS connections
Data	Session persistence	Encrypted session files
Data	API keys	Rotatable, multiple fallback keys

10.3 Secret Management



11. Technology Stack

11.1 Complete Stack Overview



11.2 Technology Details

Category	Technology	Version	Purpose
Runtime	Node.js	20+	JavaScript runtime
Database	PostgreSQL	16	Primary database
Extension	TimescaleDB	2.x	Time-series data
Extension	pgvector	0.5+	Vector similarity
Cache	Redis	7.x	Message queue & cache
Queue	BullMQ	5.x	Job processing
Browser	Playwright	1.40+	Web scraping
AI	Google Gemini	2.0 Flash	Primary AI model
AI	OpenRouter	-	Secondary AI model
AI	Ollama	-	Local AI fallback
Bot	Telegraf	4.x	Telegram bot framework
Monitoring	Prometheus	2.x	Metrics collection
Monitoring	Grafana	10.4	Visualization
Logging	Winston	3.x	Structured logging
Validation	Zod	3.x	Schema validation
Container	Docker	24+	Containerization
Orchestration	Docker Compose	2.x	Local orchestration

12. Cost Analysis

12.1 Current Infrastructure Costs

```
pie title Monthly Cost Breakdown
    "VPS (4GB RAM)" : 20
    "Gemini API" : 10
    "Domain" : 1
    "Telegram" : 0
```

Component	Provider	Monthly Cost	Notes
VPS (4GB)	Hetzner/DigitalOcean	$20	Single instance
Gemini API	Google	$5-15	Variable usage
OpenRouter	OpenRouter	$0-5	Fallback usage
Telegram Bot	Telegram	Free	No cost
Domain	Namecheap/Cloudflare	$1	Optional
Monitoring	Self-hosted	$0	Prometheus/Grafana
Total		**$26-36**	

12.2 Projected Scaling Costs

Phase	Infrastructure	Monthly Cost	Capacity
Phase 1 (Current)	Single VPS	$26-36	1,200 deals/day
Phase 2 (K8s)	3-node cluster	$150-200	10,000 deals/day
Phase 3 (Multi-region)	2 regions	$400-600	50,000 deals/day

12.3 Cost Optimization Strategies

1. **AI Model Routing**: Use cheaper models for simple tasks
2. **Caching**: Redis caching reduces API calls
3. **Batch Processing**: Process deals in batches where possible
4. **Smart Scheduling**: Post during high-engagement hours only

5. **Image Optimization**: Compress images before posting

Appendix A: File Structure

```
affiliate-platform/
├── agents/
|   ├── affiliate-system/
|   ├── curated-lists/
|   ├── deal-analysis/
|   ├── deal-discovery/
|   ├── quality-control/
|   ├── scheduler/
|   ├── telegram-posting/
|   └── whatsapp-ingestion/
├── shared/
|   ├── models/
|   ├── types/
|   └── utils/
├── infrastructure/
|   ├── db/
|   |   └── init/
|   ├── docker/
|   ├── grafana/
|   |   ├── dashboards/
|   |   └── datasources/
|   ├── monitoring/
|   └── prometheus/
├── scripts/
├── tests/
├── docker-compose.yml
└── .env
```

Appendix B: API Integration Points

Service	Endpoint	Purpose	Rate Limit
Google Gemini	generativelanguage.googleapis.com	Deal scoring	60 RPM
OpenRouter	openrouter.ai/api	Fallback AI	20 RPM
Telegram	api.telegram.org	Bot API	30 MSG/s
Amazon India	amazon.in	Deal discovery	N/A (scraping)

Appendix C: Operational Runbooks

Restarting Services

```
docker compose restart [service-name]
```

Viewing Logs

```
docker compose logs -f [service-name]
```

Database Backup

```
docker compose exec postgres pg_dump -U affiliate_user affiliate_platform > backup.sql
```

Redis Queue Inspection

```
docker compose exec redis redis-cli LLEN bull:deals:wait
docker compose exec redis redis-cli LLEN bull:posts:wait
```

Document Control

Version	Date	Author	Changes
1.0	Nov 2024	System	Initial architecture
2.0	Dec 2026	System	Added curated lists, A/B testing, content strategy

Review Schedule: Quarterly

Distribution: Internal Engineering Team

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